Sorting Trays After New Labels

AP Statistics · Unit 3 · Topic 3.7 · B: 2026-12-14 · E: 2026-12-16 · TEACHER KEY

CED TETHER

- **Skill 4.E** — Justify a claim using a decision based on significance tests.
- **EU DAT-3** — Significance testing allows us to make decisions about hypotheses within a particular context.
- **LO DAT-3.B** — Justify a claim about the population based on the results of a significance test for a population proportion. [Skill 4.E]
- **EK DAT-3.B.1** — The significance level, α, is the predetermined probability of rejecting the null hypothesis given that it is true.
- **EK DAT-3.B.2** — A formal decision explicitly compares the p-value to the significance level, α. If the p-value $\leq \alpha$, reject the null hypothesis. If the p-value $> \alpha$, fail to reject the null hypothesis.
- **EK DAT-3.B.3** — Rejecting the null hypothesis means there is sufficient statistical evidence to support the alternative hypothesis. Failing to reject the null means there is insufficient statistical evidence to support the alternative hypothesis.
- **EK DAT-3.B.4** — The conclusion about the alternative hypothesis must be stated in context.
- **EK DAT-3.B.5** — A significance test can lead to rejecting or not rejecting the null hypothesis, but can never lead to concluding or proving that the null hypothesis is true. Lack of statistical evidence for the alternative hypothesis is not the same as evidence for the null hypothesis.
- **EK DAT-3.B.6** — Small p-values indicate that the observed value of the test statistic would be unusual if the null hypothesis and probability model were true, and so provide evidence for the alternative. The lower the p-value, the more convincing the statistical evidence for the alternative hypothesis.
- **EK DAT-3.B.7** — p-values that are not small indicate that the observed value of the test statistic would not be unusual if the null hypothesis and probability model were true, so do not provide convincing statistical evidence for the alternative hypothesis nor do they provide evidence that the null hypothesis is true.
- **EK DAT-3.B.8** — A formal decision explicitly compares the p-value to the significance α. If the p-value $\leq \alpha$, then reject the null hypothesis, H_0: p $= p_0$. If the p-value $> \alpha$, then fail to reject the null hypothesis.
- **EK DAT-3.B.9** — The results of a significance test for a population proportion can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

Essential question: How do the p-value, practical importance, and study design jointly limit a test conclusion?

DOK-3 skill: 4.E · **FRQ pattern:** `statistical-decision-vs-practical-size-and-scope`

DOK rationale: Part (c) coordinates the formal evidence decision, practical importance, and study design to repair two competing conclusions and state a justified population claim.

First take → rules + (a)–(c) → turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Record one trusted clause from each statement.
- At minute 5, read the rules box and start (a). At minute 33, ask students to separate the test decision, size of the difference, and scope of the conclusion.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Mark one defensible and one questionable clause.

- Read the rules box, work (a)–(c), revise the first take, and turn in the sheet.

Questions to ask

- What does 0.0354 assume about the September sorting rate?
- Which numbers govern the decision and describe the effect?
- What population and causal comparison does the design support?

Adult role

Solo teacher. Circulate 5 pairs. Require separate decision, importance, and scope statements.

[WARN] LOOK FOR

- Reporting 0.0354 as the probability that the director's claim is true.
- Letting a small practical gap reverse the predetermined significance-test rule.
- Generalizing a September sample to every month or attributing the result to the labels.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Uses $0.0354<0.05$ to reject and concludes below 60 percent for September trays.
- **judgment-2** (required) — Explains that practical importance cannot reverse the significance decision or establish the null.
- **judgment-3** (required) — Limits inference to September and withholds label causation because there was no randomized comparison.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] TODAY'S PROBLEM — annotated

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: "The p-value is 0.0354, so reject H_0. The labels caused sorting below 60 percent for every month."

Drew: "The 5.6-point gap may be too small to matter, so it is not significant. Accept the director's claim."

The test conditions have been checked: random sample; $250 < 500$; null-expected counts 150 and 100.

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE (5 min)

Can these September trays tell us what caused the result in every month? Give one reason from the study.

Key: Not graded. Each statement mixes a defensible idea with an overreach.

Rules callout "MAKE THE DECISION, THEN BOUND THE CLAIM (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define p and state the hypotheses for checking whether correct sorting is below 60%.

Key: Let p be the proportion of all 5,000 September trays sorted correctly. Test $H_0 : p = 0.60$ versus $H_a : p < 0.60$.

DOK 2 **(b)** Calculate $\widehat{p}$, the one-proportion z statistic, and the lower-tail p-value. Use the verified conditions above.

Key: $\widehat{p} = 136/250 = 0.544$. Random is supported by the stated SRS. The 10 percent condition holds because $250 < 0.10(5{,}000) = 500$. Under H_0, $np_0 = 250(0.60) = 150$ and $n(1-p_0) = 250(0.40) = 100$, both at least 10. Thus $SE_0 = \sqrt{0.60(0.40)/250} = 0.0309839$ and $z = (0.544 - 0.60)/0.0309839 = -1.8074$. The lower-tail normal probability is $P(Z \leq -1.8074) = 0.0354$.

DOK 3 **(c)** In 4–5 sentences, correct both reports. Compare the p-value with α, separate statistical significance from practical importance, and limit the conclusion to the people or items studied and what the design can show about cause.

Key: Imani correctly rejects H_0 because $0.0354 < 0.05$: there is convincing evidence that fewer than 60% of September trays were sorted correctly. Drew can question whether the 5.6-percentage-point sample gap matters in practice, but that does not reverse the statistical decision or establish the null claim. The SRS represents September trays. Without random assignment and a comparison group, the study does not show that the labels caused the result. It also cannot establish what happens every month.

EXIT

One thing the rules box changed about my first take: ______________________________